

McMaster University
Signals and Systems

MECHTRON/SFWRENG 3MX3, Term I 2024-2025

Instructor

[Dr. M. v. Mohrenschildt](#), ITB 164, mohrens

Office hours: To be determined My main way to communicate is email. Do not sent me messages in other systems, I am not monitoring them.

Lectures, Tutorials

Lectures are in person:

- Mo 3:30PM - 4:20PM JHE 376
- Tu 4:30PM - 5:20PM ITB 137
- Th 3:30PM - 4:20PM ITB 137

Tutorials: See your Mosaic choice.

1. Mo 10:30AM - 11:20AM ETB 238
2. Tu 11:30AM - 12:20PM ABB 271
3. Fr 11:30AM - 12:20PM KTH B124
4. Fr 2:30PM - 3:20PM ABB 271

This course is an in person course (in contract to an online course), we have weekly contact hours in form of live interactive lectures/tutorials and scheduled assignments/tests throughout the term. If you have problems in participating in this live lectures/tutorials structure please inform the instructor. If you need accommodation don't hesitate to contact the instructor or Student Accommodation Services (SAS).

Notices

Updated as needed

🔒 **No tutorial the first week**

🔒 material/assignments are [here](#) (password protected)

Teaching Assistants

- Lesley Wheat, wheatd@mcmaster.ca (Monday)
- Liz Hofer, hofere1@mcmaster.ca (Tuesday)

- Howard Zhang zhanh7@mcmaster.ca (Friday 12:30)
- Ali Naqvi naqv18@mcmaster.ca (Friday 2:30)

Course Information on Web, Communication, Electronic resources

- I will make announcements by **email**
- The study material and the latest information about the course is on the **web page** (password given in first email/lecture)
- This document is located at: <http://www.cas.mcmaster.ca/~mohrens/3mx3/outline.html> .

Grades, Tests , and Exams

- There will be two midterms (quizzes) and one final.
- There will be a set of practice exercises, but no credit is given for these practice exercises. The material in the practice exercises will be tested in the Tests and Finals.
- The midterms will happen during regular scheduled class time.
- The final examination will be scheduled by the Registrars office in the usual way. It will be two hours in duration and cover the material of the course, handouts and lab-assignments. Calculators (the standard McMaster calculator) will be permitted in the tests and the final.
- Any complains about grading should be done in written and within two weeks of the return date.
- The instructor does not accept "after the fact" MSAF notices, if you write a test then the grade is yours. If you do not write a test and do not provide an acceptable MSAF notice then it is marked with 0. **The instructor reserves the right to conduct any deferred exams orally.**
- The final grade will be based on the tests and the two-hour final examination. There is one marking scheme.

Test 1, 20% , Test 2: 20%, Final 60%.

NOTE There will be no makeup tests, if you do not write a test the weight will be redistributed to futures tests.

- "The instructor and university reserve the right to modify elements of the course during the term. The university may change the dates and deadlines for any or all courses in extreme circumstances. If either type of modification becomes necessary, reasonable notice and communication with the students will be given with explanation and the opportunity to comment on changes. It is the responsibility of the student to check their McMaster email and course websites weekly during the term and to note any changes."

Book

The book is not required, but the course uses the notation from this book and covers most topics from this book.

"Structure and Interpretation of Signals and Systems", E. Lee, P. Varaiya, ISBN 0-201-74551-8
Addison Wesley

Graduate Attributes, Major Topics

- Signals: Continuous, Discrete
- Systems: Continuous, Discrete, Models, States, Differential and Difference Equations.
- Linear Time-Invariant Systems (LTI): Representation, Delay, time invariance, Impulse Response.
- Frequency Domain: Fourier Series, Complex Fourier series, Frequency Response, Convolution
- Filters: FIR Filters, IIR Filters, design and implementation
- Fourier Transforms
- Sampling Theorem: Aliasing, Nyquist-Shannon Sampling Theorem
- Laplace Transform, Stability

Applications and examples are given throughout the course.

Learning Objectives/Graduate Attribute

The Canadian Engineering Accreditation Board (CEAB) is a division of Engineers Canada and is responsible for accrediting undergraduate engineering programs across Canada. Accreditation by the CEAB ensures that the engineering programs meet a national standard of quality and cover essential educational requirements. Graduate Attributes are a set of qualities and skills that the CEAB expects engineering graduates to possess. These attributes are a benchmark for the learning outcomes of accredited engineering programs. This section lists the Graduate Attribute Indicators associated with the Learning Outcomes in this course.

- Students should know and understand
 - Period, Frequency, and normalized frequency for continuous and discrete signals
 - Discrete Linear Time Invariant systems, difference equation, state space equation, impulse response, convolution
 - The frequency domain. Periodic signals and series, Frequency response, complex exponentials Fourier transform from discrete and continuous systems.
 - Filtering, design of simple second order discrete filters.
 - The properties of transforms and applications
 - Sampling Theorem, Aliasing

- Block diagrams, feedback loop
- Stability (BIBO) of LTI systems (discrete and continuous) Laplace transform, Z transform, concepts related to signal processing and control.
- Students should be able to
 - Transform LTI systems between their different representations (difference equation, state space, impulse response, frequency response, block diagrams).
 - Compute the transform of simple signals by hand (constants, impulses, sinusoids, ...)
 - Show understanding in complex numbers, complex exponentials and their application to signals, systems, and applications such as filtering, and control
 - Model simple systems like compound interest, harmonic oscillator, simple feedback loop, ... as systems and use impulse response and transfer function to analyze them
 - Design simple digital filters by placing zeros.

NOTE:

ACADEMIC INTEGRITY

You are expected to exhibit honesty and use ethical behaviour in all aspects of the learning process. Academic credentials you earn are rooted in principles of honesty and academic integrity. **It is your responsibility to understand what constitutes academic dishonesty.**

Academic dishonesty is to knowingly act or fail to act in a way that results or could result in unearned academic credit or advantage. This behaviour can result in serious consequences, e.g. the grade of zero on an assignment, loss of credit with a notation on the transcript (notation reads: “Grade of F assigned for academic dishonesty”), and/or suspension or expulsion from the university. For information on the various types of academic dishonesty please refer to the Academic Integrity Policy, located at <https://secretariat.mcmaster.ca/university-policies-procedures-guidelines/>

The following illustrates only three forms of academic dishonesty:

- plagiarism, e.g. the submission of work that is not one’s own or for which other credit has been obtained.
- improper collaboration in group work.
- copying or using unauthorized aids in tests and examinations.

AUTHENTICITY / PLAGIARISM DETECTION

Some courses may use a web-based service (Turnitin.com) to reveal authenticity and ownership of student submitted work. For courses using such software, students will be expected to submit their work electronically either directly to Turnitin.com or via an online learning platform (e.g. A2L, etc.)

using plagiarism detection (a service supported by Turnitin.com) so it can be checked for academic dishonesty. Students who do not wish their work to be submitted through the plagiarism detection software must inform the Instructor before the assignment is due. No penalty will be assigned to a student who does not submit work to the plagiarism detection software. All submitted work is subject to normal verification that standards of academic integrity have been upheld (e.g., on-line search, other software, etc.). For more details about McMaster's use of Turnitin.com please go to www.mcmaster.ca/academicintegrity.

COURSES WITH AN ON-LINE ELEMENT

Some courses may use on-line elements (e.g. e-mail, Avenue to Learn (A2L), LearnLink, web pages, capa, Moodle, ThinkingCap, etc.). Students should be aware that, when they access the electronic components of a course using these elements, private information such as first and last names, user names for the McMaster e-mail accounts, and program affiliation may become apparent to all other students in the same course. The available information is dependent on the technology used. Continuation in a course that uses on-line elements will be deemed consent to this disclosure. If you have any questions or concerns about such disclosure please discuss this with the course instructor.

ONLINE PROCTORING

Some courses may use online proctoring software for tests and exams. This software may require students to turn on their video camera, present identification, monitor and record their computer activities, and/or lock/restrict their browser or other applications/software during tests or exams. This software may be required to be installed before the test/exam begins.

CONDUCT EXPECTATIONS

As a McMaster student, you have the right to experience, and the responsibility to demonstrate, respectful and dignified interactions within all of our living, learning and working communities. These expectations are described in the Code of Student Rights & Responsibilities (the "Code"). All students share the responsibility of maintaining a positive environment for the academic and personal growth of all McMaster community members, whether in person or online. It is essential that students be mindful of their interactions online, as the Code remains in effect in virtual learning environments. The Code applies to any interactions that adversely affect, disrupt, or interfere with reasonable participation in University activities. Student disruptions or behaviours that interfere with university functions on online platforms (e.g. use of Avenue 2 Learn, WebEx or Zoom for delivery), will be taken very seriously and will be investigated. Outcomes may include restriction or removal of the involved students' access to these platforms.

ACADEMIC ACCOMMODATION OF STUDENTS WITH DISABILITIES

Students with disabilities who require academic accommodation must contact Student Accessibility Services (SAS) at 905-525-9140 ext. 28652 or sas@mcmaster.ca to make arrangements with a Program Coordinator. For further information, consult McMaster University's Academic Accommodation of Students with Disabilities policy.

REQUESTS FOR RELIEF FOR MISSED ACADEMIC TERM WORK

McMaster Student Absence Form (MSAF): In the event of an absence for medical or other reasons, students should review and follow the Academic Regulation in the Undergraduate Calendar “Requests for Relief for Missed Academic Term Work”.

ACADEMIC ACCOMMODATION FOR RELIGIOUS, INDIGENOUS OR SPIRITUAL OBSERVANCES (RISO)

Students requiring academic accommodation based on religious, indigenous or spiritual observances should follow the procedures set out in the RISO policy. Students should submit their request to their Faculty Office normally within 10 working days of the beginning of term in which they anticipate a need for accommodation or to the Registrar's Office prior to their examinations. Students should also contact their instructors as soon as possible to make alternative arrangements for classes, assignments, and tests.

COPYRIGHT AND RECORDING

Students are advised that lectures, demonstrations, performances, and any other course material provided by an instructor include copyright protected works. The Copyright Act and copyright law protect every original literary, dramatic, musical and artistic work, including lectures by University instructors. The recording of lectures, tutorials, or other methods of instruction may occur during a course. Recording may be done by either the instructor for the purpose of authorized distribution, or by a student for the purpose of personal study. Students should be aware that their voice and/or image may be recorded by others during the class. Please speak with the instructor if this is a concern for you.

EXTREME CIRCUMSTANCES

The University reserves the right to change the dates and deadlines for any or all courses in extreme circumstances (e.g., severe weather, labour disruptions, etc.). Changes will be communicated through regular McMaster communication channels, such as McMaster Daily News, A2L and/or McMaster email.